Transfer Learning Framework for Prediction of Alzheimer's Disease

A Project Report

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in

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Under the Guidance of

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SCHOOL OF TECHNOLOGY
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THE APOLLO UNIVERSITY, CHITTOOR

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DECLARATION

I hereby declare that the project entitled "Transfer Learning Framework for Prediction of Alzheimer's Disease" submitted by us, for the award of the degree of Bachelor of Technology in Computer Science & Engineering, School of Technology to The Apollo University is a record of bonafide work carried out by us under the supervision of "Dr. G. B. Hima Bindu".

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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BONAFIDE CERTIFICATE

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ABSTRACT

Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking ability, and behavior. Early detection of Alzheimer’s disease is important for providing proper treatment and improving patient care. Traditional diagnosis methods rely on manual analysis of MRI brain images, which is time-consuming and may lead to errors. Therefore, automated prediction systems using deep learning techniques are required. This project proposes a transfer learning based deep learning framework for Alzheimer’s disease prediction using MRI brain images. The system uses pre-trained convolutional neural network models such as DenseNet121, MobileNetV2, and EfficientNetB0. These models are used for feature extraction and classification of MRI images. The dataset consists of four classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. Image preprocessing techniques such as resizing, normalization, and data augmentation are applied to improve model performance. Transfer learning helps in reducing training time and improving prediction accuracy. The trained models are evaluated using performance metrics such as accuracy, precision, recall, and F1-score. The experimental results show that the proposed system achieves high accuracy in predicting Alzheimer’s disease. The system can assist doctors in early diagnosis and improve decision making. This project demonstrates the effectiveness of deep learning transfer learning models in medical image classification.

Keywords: Alzheimer’s Disease, Transfer Learning, Deep Learning, MRI Classification, Hyperparameter Tuning

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LIST OF ABBREVIATIONS

AD: Alzheimer’s Disease

MRI: Magnetic Resonance Imaging

CNN: Convolutional Neural Network

DL: Deep Learning

TL: Transfer Learning

AI: Artificial Intelligence

ML: Machine Learning

VGG: Visual Geometry Group Network

ResNet: Residual Neural Network

DenseNet: Densely Connected Convolutional Network

EfficientNet: Efficient Convolutional Neural Network

ReLU: Rectified Linear Unit

Softmax: Softmax Activation Function

Dropout: Regularization Technique to Prevent Overfitting

Epoch: One Complete Training Cycle

Batch Size: Number of Samples per Training Step

LR: Learning Rate

GPU: Graphics Processing Unit

TP: True Positive

TN: True Negative

FP: False Positive

FN: False Negative

CHAPTER 1: INTRODUCTION

1.1. BACKGROUND OF THE STUDY

Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking ability, and behavior. It is one of the most common causes of dementia among older adults. The disease gradually damages brain cells, leading to loss of cognitive functions and difficulty in performing daily activities. Early detection of Alzheimer’s disease is very important because it helps in slowing down disease progression and improving patient care.

Medical imaging techniques such as Magnetic Resonance Imaging (MRI) are widely used to analyze structural changes in the brain. MRI scans help doctors observe brain shrinkage, tissue damage, and abnormalities associated with Alzheimer’s disease. However, manual analysis of MRI images requires expert radiologists and is time-consuming. In many cases, early symptoms are difficult to identify visually, which may lead to delayed diagnosis.

With the advancement of artificial intelligence and deep learning, automated systems can be developed for disease prediction. Deep learning models, especially Convolutional Neural Networks (CNN), are widely used for image classification tasks. These models automatically extract features from images and provide accurate predictions. However, training deep learning models from scratch requires large datasets and high computational resources.

Transfer learning is an effective technique that uses pre-trained deep learning models for new tasks. These models are already trained on large datasets and can be fine-tuned for medical image classification. Transfer learning reduces training time and improves prediction accuracy. Models such as DenseNet121, MobileNetV2, and EfficientNetB0 are widely used for medical image analysis.

In this project, a transfer learning based deep learning framework is proposed for Alzheimer’s disease prediction using MRI images. The system classifies images into four categories: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. The proposed system helps in early detection and assists medical professionals in decision making.

1.2. PROBLEM STATEMENT

Alzheimer's disease is difficult to diagnose in its early stages because the symptoms develop gradually over time. Traditional diagnosis methods rely on manual analysis of MRI brain images by radiologists. This process is time-consuming, subjective, and prone to human errors. In addition, the increasing number of patients makes manual diagnosis challenging.

Existing machine learning methods require manual feature extraction, which reduces accuracy and increases complexity. These methods also fail to capture complex patterns in MRI images. Deep learning models provide better performance, but training them from scratch requires large datasets and high computational cost.

There is a need for an automated and efficient system that can accurately predict Alzheimer's disease using MRI images. The system should reduce human effort, improve diagnosis speed, and provide high accuracy. Transfer learning based deep learning models can solve this problem by using pre-trained architectures and fine-tuning them for Alzheimer's disease prediction.

Therefore, this project proposes a transfer learning based deep learning framework using DenseNet121, MobileNetV2, and EfficientNetB0 models. The system automatically extracts features from MRI images and classifies them into different Alzheimer's disease categories. This approach improves prediction accuracy and supports early diagnosis.

CHAPTER 2: LITERATURE REVIEW

2.1. INTRODUCTION

System analysis is an important phase in software development that involves understanding the current system, identifying its limitations, and proposing an improved solution. It helps in defining system requirements, evaluating existing processes, and designing a system that meets user needs effectively.

In the context of the **Alzheimer’s Disease Prediction System**, system analysis focuses on studying the current methods used for diagnosing Alzheimer’s disease and identifying the challenges faced in early prediction. Alzheimer’s disease is a progressive neurological disorder that affects memory, thinking ability, and behavior. Early detection plays a crucial role in providing proper treatment and slowing disease progression.

This project provides a detailed analysis of the existing Alzheimer’s disease detection methods, highlights their limitations, and presents an overview of the proposed deep learning-based prediction system. The analysis helps in understanding how the new system improves prediction accuracy, reduces manual effort, and provides faster diagnosis.

2.2. REVIEW OF EXISTING SYSTEM

The existing Alzheimer’s disease detection system mainly relies on clinical diagnosis and manual analysis of MRI brain images. Doctors and radiologists examine MRI scans visually to identify structural changes in the brain such as shrinkage and tissue loss. This process requires expert knowledge and takes significant time.

Some traditional machine learning-based systems are also used for Alzheimer’s prediction. These systems involve manual preprocessing of MRI images followed by feature extraction. Features such as texture, shape, and intensity are extracted and fed into classifiers like Support Vector Machine (SVM), K-Nearest Neighbors (KNN), or Random Forest for prediction.

However, these systems depend heavily on manual feature extraction and domain expertise. The performance of the model depends on the quality of features selected. This makes the prediction process complex and less accurate.

Additionally, many existing systems are not fully automated. They require human intervention at multiple stages including preprocessing, feature extraction, and classification. This increases processing time and reduces efficiency.

Moreover, some systems use small datasets which results in overfitting and poor generalization. These limitations highlight the need for a deep learning-based automated system for Alzheimer's disease prediction.

Table 2.1 - Summary of Literature Review

S.No	Author & Year	System	Key Contribution	Limitation
1	Zhang et al. (2022)	CNN Model	Automatic MRI classification	Limited dataset
2	Liu & Wang (2023)	SVM Based System	Feature-based prediction	Manual feature extraction
3	Patel et al. (2023)	Transfer Learning Model	Pretrained CNN usage	Moderate accuracy
4	Sharma & Gupta (2024)	Deep CNN Model	Improved classification	High training time
5	Kumar et al. (2024)	Hybrid Deep Model	Better feature learning	Computational cost

6	WHO Report (2024)	Alzheimer Study	Importance of early detection	No automated solution

2.3. LIMITATIONS OF EXISTING SYSTEM

The existing Alzheimer's disease detection systems have several limitations. One major drawback is the dependency on manual diagnosis. Radiologists must analyze MRI images manually, which is time-consuming and prone to human error.

Another limitation is the use of traditional machine learning algorithms. These methods require manual feature extraction, which is complex and may not capture hidden patterns in MRI images. This affects the accuracy of prediction.

The existing systems also suffer from low prediction accuracy. Early-stage Alzheimer's disease is difficult to detect using traditional methods. This reduces the effectiveness of treatment planning.

Data handling is another issue. Some systems cannot efficiently process large MRI datasets. Training models on large datasets requires high computational resources, which is not supported in existing approaches.

Overfitting is also a common problem. Models trained on small datasets may perform well on training data but fail on new test data. This reduces reliability.

The existing systems also lack automation. Most systems require manual preprocessing and intervention, making the process slow and inefficient.

Overall, the existing system is time-consuming, less accurate, and lacks automation.

2.4. PROPOSED SYSTEM OVERVIEW

To overcome the limitations of the existing system, an **Alzheimer's Disease Prediction System using Transfer Learning** is proposed. The proposed system uses deep learning models to automatically classify MRI brain images.

In the proposed system, MRI images are collected and preprocessed. The dataset is divided into training, validation, and testing sets. Transfer learning models such as DenseNet, MobileNet, and EfficientNet are used for feature extraction and classification.

The deep learning model automatically learns features from MRI images without manual feature extraction. The trained model predicts the stage of Alzheimer's disease. The system classifies images into categories such as Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented.

The proposed system improves prediction accuracy and reduces manual effort. The model training is optimized using parameters such as learning rate, dropout rate, and dense layers. Validation accuracy is used to monitor model performance.

The system also provides performance evaluation using accuracy, loss, confusion matrix, and classification report. These metrics help in analyzing the effectiveness of the model.

Overall, the proposed system provides an automated, accurate, and efficient solution for Alzheimer's disease prediction using deep learning techniques.

Table 2.2 - comparison of existing system and proposed system

Feature	**Existing System**	**Proposed System**
Detection Method	Manual, Traditional ML	Deep Learning (Transfer Learning)
Feature Extraction	Manual	Automatic
Accuracy	Moderate	High
Processing Time	Slow	Fast
Automation	Partial	Fully Automated

CHAPTER 3: OBJECTIVES AND SCOPE OF WORK

3.1. OBJECTIVES OF THE PROJECT

The primary objective of the Alzheimer's Disease Prediction System is to design and develop a deep learning-based model that can automatically detect Alzheimer's disease using MRI brain images. The system aims to assist medical professionals in early diagnosis by providing accurate and reliable predictions.

One of the key objectives of the project is to develop an automated prediction system using transfer learning techniques. The system uses pretrained deep learning models to extract features from MRI images and classify them into different stages of Alzheimer's disease. This reduces the need for manual feature extraction and improves prediction accuracy.

Another important objective is to improve early detection of Alzheimer's disease. Early diagnosis helps doctors provide timely treatment and slow the progression of the disease. The system aims to classify MRI images into categories such as Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

The project also aims to reduce human effort in diagnosis. In traditional systems, radiologists manually examine MRI images, which is time-consuming and prone to error. The proposed system automates the classification process and provides faster results.

Ensuring high prediction accuracy is another major objective. The system uses transfer learning models such as DenseNet, MobileNet, and EfficientNet to improve performance. Hyperparameter tuning is applied to optimize learning rate, dropout rate, and dense layers.

The project also focuses on performance evaluation. Metrics such as accuracy, validation accuracy, loss, confusion matrix, and classification report are used to analyze model performance. These metrics help in understanding the effectiveness of the model.

Another objective is to build a scalable model that can handle large datasets. The system is designed to train on multiple MRI images and provide consistent predictions.

Overall, the objectives of the project focus on improving prediction accuracy, enabling early detection, reducing manual effort, and providing an automated solution for Alzheimer’s disease classification.

3.2. SCOPE OF THE PROJECT

The scope of the Alzheimer’s Disease Prediction System includes the design, development, and implementation of a deep learning model for classifying MRI brain images. The system focuses on detecting Alzheimer’s disease at different stages using transfer learning techniques.

The project involves collecting MRI brain image datasets and preprocessing them before training. The dataset is divided into training, validation, and testing sets. This helps in building an efficient and reliable prediction model.

The system includes training deep learning models such as DenseNet, MobileNet, and EfficientNet. These models are used for feature extraction and classification. The trained model predicts the stage of Alzheimer’s disease based on the input MRI image.

Another part of the scope includes model evaluation. The system evaluates performance using metrics such as accuracy, loss, confusion matrix, and classification report. These results help in comparing model performance and selecting the best model.

The project also includes visualization of training results. Graphs such as training accuracy, validation accuracy, training loss, and validation loss are generated. These graphs help in analyzing model learning.

The scope also covers prediction of new MRI images. The trained model can take new MRI input and classify it into appropriate Alzheimer’s categories. This helps in real-time prediction.

However, certain functionalities are outside the scope of the project. The system does not include real-time hospital integration, patient medical records, or treatment recommendation. The project focuses only on prediction using MRI images.

The system focuses on developing a scalable deep learning model for Alzheimer’s disease prediction using MRI images.

3.3. GANTT CHART

In the Alzheimer's Disease Prediction project, the development was carried out in multiple phases including dataset collection, preprocessing, model selection, training, testing, and documentation.

The following table represents the schedule of work carried out during the project:

Table 3.1 - Gantt Chart

TASK	MONTH 1	MONTH 2	MONTH 3	MONTH 4
Topic Finalization	✓			
Literature Review	✓	✓		
Dataset Collection	✓	✓		
Data Preprocessing		✓		
Model Selection		✓		
Model Training			✓	
Testing & Validation			✓	✓
Performance Evaluation			✓	✓
Documentation				✓
Final Submission				✓

CHAPTER 4: SYSTEM ANALYSIS & DESIGN

4.1. SYSTEM REQUIREMENTS (HARDWARE & SOFTWARE)

The successful implementation of the Alzheimer's Disease Prediction System requires specific hardware and software resources. These requirements ensure smooth model training, testing, and prediction.

4.1.1. Hardware Requirements

The hardware requirements for developing and running the deep learning model are moderate. A system with good RAM and processor is recommended for training the model efficiently.

Table 4.1 - Hardware Requirements

Component	**Specification**
Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB recommended)
Storage	500 GB or above
System Type	64-bit OS
GPU	Cloud GPU (Google Colab)
Output Devices	Monitor

These specifications are sufficient for model training and prediction.

4.1.2. Software Requirements

The software requirements include tools and technologies used for building the Alzheimer's prediction model.

Table 4.2 - Software Requirements

Category	Tools / Technologies
Operating System	Windows / Linux / macOS
Programming Language	Python
Framework	TensorFlow, Keras
Libraries	NumPy, Scikit-learn
Visualization	Matplotlib, Seaborn
Models	DenseNet121, MobileNetV2, EfficientNetB0
Platform	Google Colab
Dataset	MRI Brain Images

These technologies help in building an accurate and efficient prediction system.

4.2. SYSTEM ARCHITECTURE

The Alzheimer's Disease Prediction System is designed using a layered architecture, which ensures modularity, maintainability, and better performance. The architecture is divided into multiple layers: Input Layer, Preprocessing Layer, Model Layer, Training

& Fine-Tuning Layer, and Output Layer. Each layer performs a specific set of tasks, which collectively contribute to accurate prediction of Alzheimer's disease stages from MRI brain images.

4.2.1 Input Layer

The Input Layer is responsible for accepting MRI brain images from the dataset. The dataset used in this project consists of labeled MRI scans categorized into four Alzheimer's disease stages:

1. Non-Demented
2. Very Mild Demented
3. Mild Demented
4. Moderate Demented

These images are used for both training the deep learning models and testing their performance. Proper organization and labeling of the dataset are essential to ensure that the models learn the features accurately. Additionally, input images are validated to check for proper size, format, and quality before further processing.

4.2.2 Preprocessing Layer

The Preprocessing Layer prepares MRI images for effective model training and helps reduce overfitting. This layer performs the following steps:

- **Image Resizing:** All MRI images are resized to 224×224 pixels to maintain uniform input size across models.
- **Normalization:** Pixel values are normalized to scale the data between 0 and 1. This improves training stability and convergence.
- **Data Augmentation:** Techniques such as rotation**,** horizontal and vertical flipping, and zooming are applied to artificially increase dataset size. This helps the model generalize better and reduces the chance of overfitting.
- **Dataset Split:** Images are split into training and validation/test sets to evaluate model performance effectively.

This preprocessing ensures that the input data is clean, uniform, and enriched for robust feature extraction.

4.2.3 Model Layer

The Model Layer uses transfer learning, which leverages pretrained deep learning models to extract features from MRI images efficiently. The models used in this project include:

- DenseNet121
- MobileNetV2
- EfficientNetB0

These models are pretrained on the ImageNet dataset and can extract hierarchical features from images. The base layers of these models are initially frozen to retain the pretrained features. On top of the base model, custom classification layers are added, including Dense layers, Dropout layers, and a Softmax output layer. This allows the models to learn domain-specific features related to Alzheimer's disease while benefiting from the pretrained weights.

4.2.4 Training and Fine-Tuning Layer

The model training is performed in **two stages**:

1. **Frozen Layer Training:** Only the newly added classification layers are trained while keeping the base layers frozen. This ensures that the pretrained features are not disturbed.
2. **Fine-Tuning:** The last few layers of the base model are unfrozen, and the entire network is trained with a low learning rate to refine feature extraction for MRI images.

Additionally, hyperparameter tuning is performed to optimize model performance. Key hyperparameters include:

- Learning rate
- Dropout rate

- Batch size
- Number of epochs

This process ensures the model achieves the best balance between accuracy and generalization.

4.2.5 Output Layer

The Output Layer performs multi-class classification using a Softmax activation function. The model predicts one of the following categories for each MRI image:

- Non Demented
- Very Mild Demented
- Mild Demented
- Moderate Demented

Performance metrics such as accuracy, validation loss, and confusion matrix are generated to evaluate the model's effectiveness. Correct predictions and misclassifications are analyzed to improve model design further.

Architecture Flow

The architecture flow of the system can be summarized as follows:

1. **Dataset Collection:** MRI brain images are collected and labeled according to Alzheimer's stages.
2. **Image Preprocessing:** Images are resized to 224×224 pixels, normalized, and augmented.
3. **Transfer Learning Model Loading:** Pretrained models are loaded for feature extraction.
4. **Frozen Layer Training:** Initial training is performed on the added classification layers.
5. **Fine-Tuning:** Selected base layers are unfrozen and trained for improved performance.
6. **Hyperparameter Tuning:** Parameters such as learning rate, dropout, and batch size are optimized.

7. **Model Evaluation:** Accuracy, loss curves, and confusion matrix are calculated for performance assessment.
8. **Prediction:** Trained models predict Alzheimer's disease stages for unseen MRI images.

Advantages of Layered Architecture

- **Modularity:** Each layer has a specific role, making the system easier to maintain and update.
- **Scalability:** Layers can be extended with new models or preprocessing techniques.
- **Security:** Input and output handling is separated from processing, reducing errors and improving safety.
- **Performance:** Layer separation allows optimization at each stage, ensuring faster training and accurate predictions.
- **Maintainability:** The modular approach simplifies debugging and future enhancements.

This layered system architecture ensures a structured, efficient, and reliable approach for predicting Alzheimer's disease stages from MRI brain images, making it suitable for real-world healthcare applications.

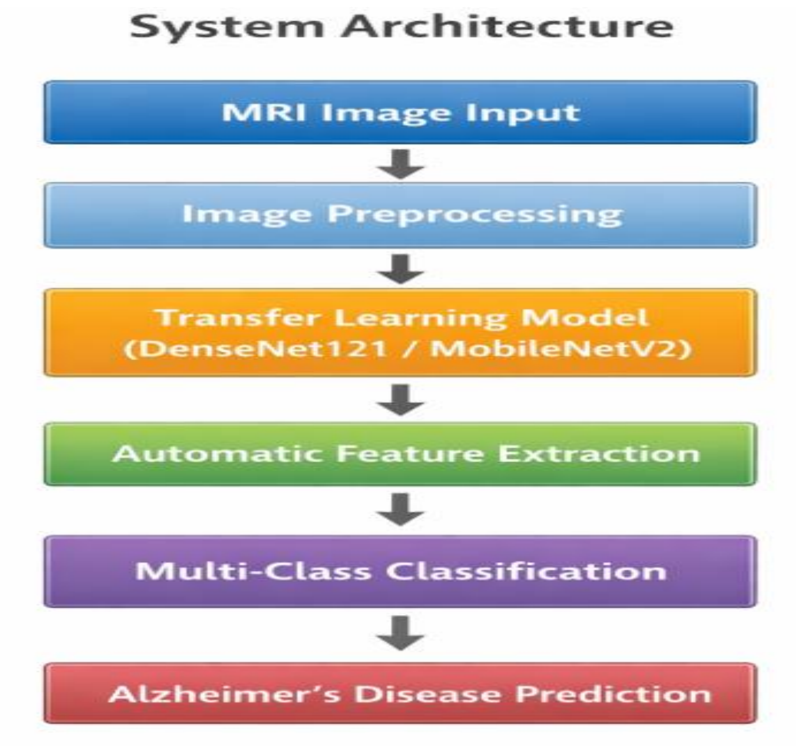

Figure 4.1:- System Architecture

4.3. DATA FLOW DIAGRAM (DFD)

4.3.1. Level 0 DFD (Context Diagram)

- **Entities**:
  - User
- **Process**:
  - Alzheimer's Disease Prediction System
- **Data Flow**:
  - User uploads MRI image → System
  - System preprocesses image → Model
  - Model predicts class → System
  - System displays result → User

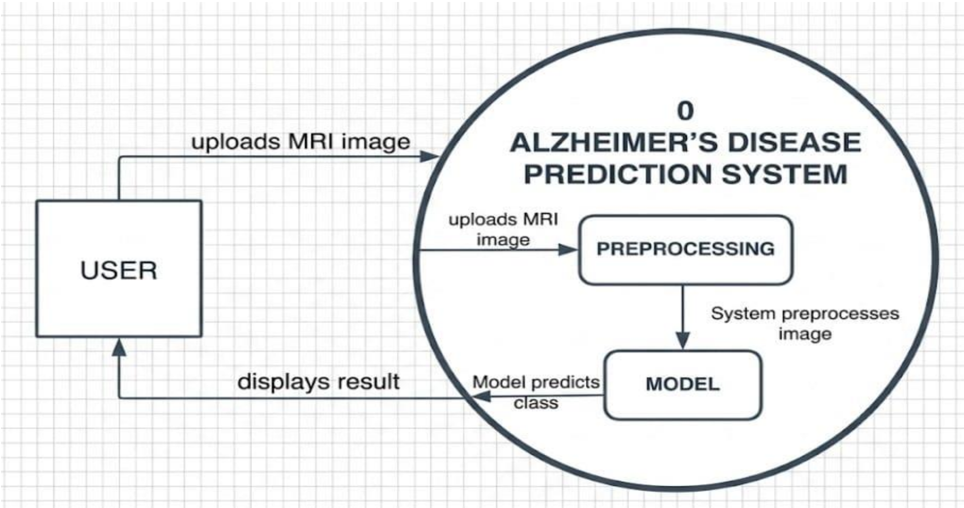

Figure 4.2 :- Level 0 DFD- Context Diagram

4.3.2. Level 1 DFD

The Level 1 DFD breaks down the system into major processes:

1. Dataset Collection
   - MRI images collected from dataset
2. Image Preprocessing
   - Resize images and apply normalization
3. Data Augmentation
   - Rotation, flipping, zooming applied
4. Transfer Learning Model
   - DenseNet121, MobileNetV2, EfficientNetB0
5. Model Training
   - Frozen layer training performed
6. Fine Tuning
   - Last layers unfrozen and trained
7. Prediction
   - Model predicts Alzheimer stage
8. Result Display
   - Accuracy and confusion matrix shown

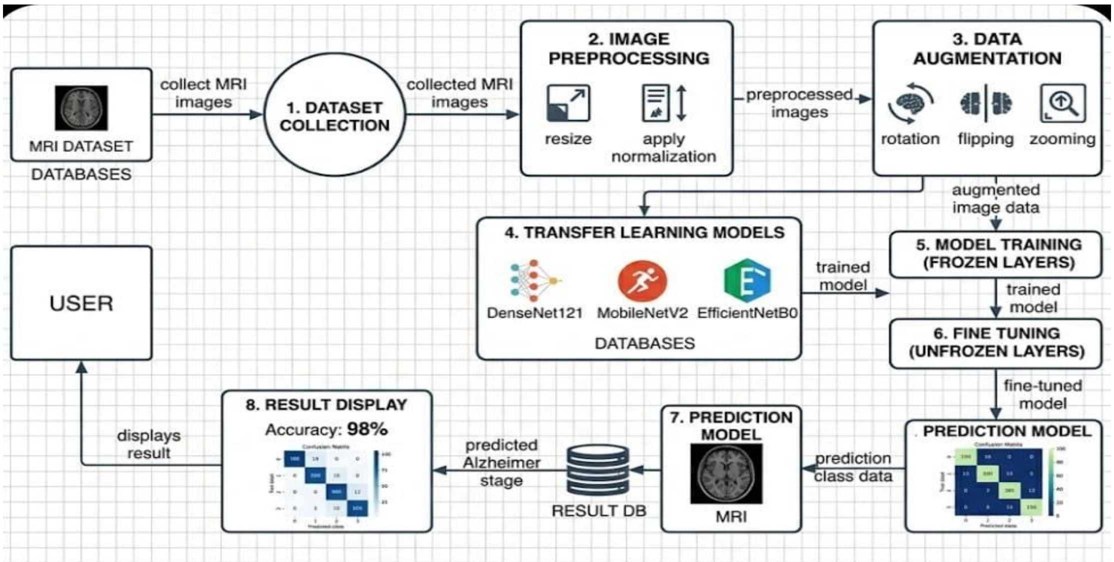

Figure 4.3 :- Level 1 Data Flow Diagram

4.3.3. Data Stores

- Dataset – Stores MRI images
- Model – Stores trained weights
- Prediction Results – Stores classification output

4.3.4. Data Flow Explanation

The MRI dataset is provided as input to the system. The images are preprocessed and resized before training. Transfer learning models extract features and perform classification. The trained model predicts Alzheimer's disease stage. The prediction results are displayed along with accuracy and confusion matrix.

4.3.5. Dataset Information Table

Table 4.3 - Alzheimer's MRI Dataset Description

Parameter	**Description**
Dataset Type	MRI Brain Images

Classes	Non-Demented, Very Mild Demented, Mild Demented, Moderate Demented
Image Size	224 × 224 pixels
Color Format	RGB
Dataset Split	Training, Validation, Testing
Source	Public Alzheimer MRI Dataset

Table 4.4 - Transfer Learning Model Configuration Table

Parameter	**Value**
Models Used	DenseNet121, MobileNetV2, EfficientNetB0
Transfer Learning	ImageNet Pretrained Weights
Input Shape	224 × 224 × 3
Optimizer	Adam
Loss Function	Categorical Crossentropy
Activation Function	Softmax
Batch Size	32
Epochs	20–30

Table 4.5 - Training HyperParameters Table

Parameter	**Value**
Learning Rate	0.0001
Dropout Rate	0.4 / 0.5
Dense Units	128 / 256
Fine Tuning	Last layers unfrozen
Data Augmentation	Rotation, Flip, Zoom
Evaluation Metrics	Accuracy, Loss
Validation Method	Validation Split

CHAPTER 5: IMPLEMENTATION

5.1 TOOLS & TECHNOLOGIES USED

The Alzheimer's Disease Prediction System is developed using deep learning and transfer learning techniques to classify MRI brain images into different stages of Alzheimer's disease. The system is implemented using Python-based frameworks and libraries that support image processing, model training, and evaluation. A GPU-enabled cloud environment is used to accelerate model training and improve computational efficiency. The selected tools and technologies provide flexibility, high performance, and better accuracy for medical image classification tasks. The following tools and technologies were used in this project.

5.1.1 Programming Language

• Python: Python is used as the primary programming language for implementing the Alzheimer's disease prediction system. Python provides extensive support for deep learning, machine learning, and image processing. It is easy to understand, flexible, and supports multiple libraries required for data preprocessing, model training, evaluation, and prediction. Python is used for loading MRI images, performing preprocessing operations, building transfer learning models, training the network, and predicting Alzheimer's disease stages.

5.1.2 Deep Learning Frameworks

● TensorFlow: TensorFlow is an open-source deep learning framework used for building and training neural network models. It provides support for transfer learning, GPU acceleration, and automatic differentiation. TensorFlow is used to implement pretrained convolutional neural network models and perform training on MRI brain images. It also helps in optimizing model performance and handling large datasets efficiently.

● Keras: Keras is a high-level deep learning API built on top of TensorFlow. It simplifies the implementation of deep learning models and provides easy-to-use functions for building neural networks. Keras is used for loading pretrained models such as DenseNet121, MobileNetV2, EfficientNetB0, VGG16, and ResNet50. It is also used for adding custom classification layers, dropout layers, compiling the model, and training the network.

5.1.3 Machine Learning Libraries

● Scikit-learn: Scikit-learn is used for evaluating the performance of the trained model. It provides functions for generating confusion matrix, classification report, and accuracy score. These metrics help in analyzing the prediction performance of the Alzheimer's disease classification system.

● NumPy: NumPy is used for performing numerical operations and handling multi-dimensional arrays. It is mainly used for processing image data, converting predictions, calculating metrics, and performing mathematical operations during model training and prediction.

● Pandas: Pandas is used for dataset organization and data handling. It helps in managing labels, storing results, and analyzing prediction outputs. Pandas also assists in creating tables for evaluation results.

5.1.4 Visualization Tools

• Matplotlib: Matplotlib is used for plotting training and validation accuracy graphs. It is also used to visualize model loss curves and performance comparison between different transfer learning models. These graphs help in understanding model learning behavior.

• Seaborn: Seaborn is used for visualizing confusion matrix and classification performance. It provides better graphical representation for analyzing prediction results. Seaborn helps in identifying correctly classified and misclassified MRI images.

5.1.5 Development Environment

● Google Colab: Google Colab is used as the primary development environment for training deep learning models. It provides free GPU support which helps in faster model training. Google Colab also supports Python libraries such as TensorFlow, Keras, NumPy, and Matplotlib. It allows easy dataset upload, model training, and result visualization.

● Jupyter Notebook: Jupyter Notebook is used for experimentation, testing, and debugging the code. It allows step-by-step execution of Python code and visualization of outputs. It is helpful for testing preprocessing steps, model building, and prediction results.

These tools and technologies collectively help in building an efficient, accurate, and reliable Alzheimer’s disease prediction system.

The Alzheimer’s Disease Prediction System is developed using deep learning and transfer learning techniques to classify MRI brain images. The implementation uses Python-based frameworks and GPU-enabled cloud environment for improving performance and accuracy. The following tools and technologies were used in this project.

5.2. MODULE DESCRIPTION

The system is divided into several modules to improve performance and maintainability.

5.2.1. Dataset Collection Module

This module collects MRI brain images from publicly available Alzheimer’s dataset. The dataset consists of multiple categories such as Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented. These images are used for training and testing the deep learning models.

5.2.2. Image Preprocessing Module

This module performs preprocessing operations on the MRI brain images before feeding them into the deep learning model. Image preprocessing is an important step to improve the model performance and ensure uniform input size. The MRI images collected from the dataset may have different sizes, orientations, and noise levels. Therefore, preprocessing helps to standardize the images and make them suitable for training.

The preprocessing steps include:

• Image resizing to 224 × 224 pixels so that all images match the input size required by transfer learning models
• Image normalization to scale pixel values between 0 and 1, which helps in faster convergence
• Data augmentation techniques such as rotation, horizontal flip, zoom, and shift to increase dataset diversity
• Train and validation split to divide dataset into training data and validation data
• Removal of noise and improvement of image quality

This preprocessing step improves model accuracy, reduces overfitting, and enhances generalization performance.

5.2.3 Transfer Learning Module

This module loads pretrained deep learning models that are trained on large-scale datasets such as ImageNet. Transfer learning helps in reducing training time and improves performance when working with limited medical image datasets. The pretrained models are used as feature extractors to capture important patterns from MRI images.

The models used in this project include:

• DenseNet121
• MobileNetV2
• EfficientNetB0
• VGG16
• ResNet50

These models extract deep features such as texture, shape, and structural abnormalities from MRI brain images. The final classification layers of these pretrained models are removed and replaced with custom layers suitable for Alzheimer's disease classification. Transfer learning helps in achieving higher accuracy with less training data.

5.2.4 Model Training Module

This module trains the transfer learning models using the preprocessed MRI dataset. During training, the pretrained base model is combined with custom classification layers. The training process adjusts weights in the classification layers to classify Alzheimer's disease stages.

The training steps include:

• Freeze base layers of pretrained model to retain learned features
• Add dense fully connected layer for feature learning
• Add dropout layer to prevent overfitting
• Add softmax output layer for multi-class classification
• Compile model using optimizer, loss function, and accuracy metric
• Train model using training dataset
• Validate model using validation dataset

The model learns important patterns from MRI images and improves classification performance over multiple epochs.

5.2.5 Hyperparameter Tuning Module

This module improves model performance by adjusting different hyperparameters. Hyperparameter tuning helps in finding the best configuration for achieving higher accuracy and lower loss. Different combinations of parameters are tested and the best model is selected.

The tuning parameters include:

• Learning rate tuning to control weight update speed
• Dropout tuning to reduce overfitting
• Dense units tuning to adjust number of neurons
• Epoch tuning to control training iterations
• Batch size tuning to control number of samples per training step
• Optimizer tuning such as Adam or RMSprop

Hyperparameter tuning helps in improving model accuracy, reducing training loss, and achieving better generalization.

5.2.6 Prediction Module

This module predicts the Alzheimer's disease stage using the trained deep learning model. The input MRI image is passed through preprocessing and then fed into the trained model. The model extracts features and classifies the image into different Alzheimer's categories.

The prediction steps include:

- Input MRI brain image
- Image preprocessing and resizing
- Feature extraction using pretrained CNN model
- Classification using softmax layer
- Output predicted class label

The model predicts one of the following classes:

- Non Demented
- Very Mild Demented
- Mild Demented
- Moderate Demented

The prediction module provides the final output indicating the Alzheimer's disease stage.

5.3. ALGORITHMS / METHODS USED

The proposed system uses deep learning-based transfer learning models to predict Alzheimer's disease stages from MRI brain images. The system is designed to automatically analyze brain MRI scans and classify them into different stages of Alzheimer's disease. Transfer learning models are used in this project because they are pretrained on large datasets and can extract important features such as texture, shape, and structural abnormalities from medical images.

Several algorithms and methods are implemented in this system, including image preprocessing, transfer learning, model training, hyperparameter tuning, and prediction. In the image preprocessing stage, MRI images are resized, normalized, and augmented to improve data quality and increase dataset diversity. This helps the model learn better features and reduces the risk of overfitting.

In the transfer learning stage, pretrained models such as DenseNet121, MobileNetV2, EfficientNetB0, VGG16, and ResNet50 are used as feature extractors. The final layers of these models are modified to perform Alzheimer's disease classification. During model training, the base layers are frozen and custom dense layers with dropout are added. The models are trained using training data and validated using validation data.

Hyperparameter tuning is performed to improve model performance by adjusting learning rate, batch size, epochs, dropout rate, and number of dense units. Different combinations are tested to obtain the best accuracy. Finally, the prediction module takes MRI images as input and outputs the predicted Alzheimer's disease stage.

5.3.1. Transfer Learning Algorithm

Transfer learning is used in this project to utilize pretrained convolutional neural network models trained on large datasets. Instead of training the model from scratch, pretrained models such as DenseNet121, MobileNetV2, EfficientNetB0, VGG16, and ResNet50 are used. These models extract important features from MRI brain images and classify Alzheimer's disease stages.

Steps involved in Transfer Learning Algorithm:

- Load pretrained CNN model with ImageNet weights
- Freeze base layers and add custom classification layers
- Apply dropout and softmax output layer
- Compile, train, and evaluate the model

This method reduces training time and improves model accuracy.

5.3.2. Image Preprocessing Algorithm

Image preprocessing is performed to improve the quality of MRI images and prepare them for model training. The preprocessing step ensures that all images have the same size and format. Data augmentation is also applied to increase dataset diversity and avoid overfitting.

Steps involved in Image Preprocessing Algorithm:

● Load MRI dataset and resize images to 224×224
● Normalize pixel values and convert format
● Apply data augmentation techniques
● Split dataset into training and validation sets

This preprocessing step improves model generalization.

5.3.3. Model Training Algorithm

Model training is performed using transfer learning models. The pretrained CNN models are trained using MRI dataset to classify Alzheimer's disease stages.

Steps involved in Model Training Algorithm:

● Load pretrained model and freeze base layers
● Add dense, dropout, and softmax layers
● Compile model and train using dataset
● Validate performance and save best model

This step trains the deep learning model for classification.

5.3.4. Hyperparameter Tuning Method

Hyperparameter tuning is performed to improve model performance. Different parameters such as learning rate, dropout rate, dense units, and epochs are adjusted. Multiple models are trained using different combinations, and the best parameters are selected based on validation accuracy.

Steps involved in Hyperparameter Tuning:

- Define learning rate, dropout, and dense units
- Train models with different parameter combinations

- Evaluate validation accuracy and compare results
- Select best hyperparameters and retrain model

Hyperparameter tuning improves accuracy and reduces overfitting.

5.3.5. Prediction Method

Prediction method is used to classify Alzheimer's disease stage from input MRI image. The trained model is used to predict the output class.

Steps involved in Prediction Method:

● Load trained model and input MRI image
● Preprocess image and extract features
● Apply softmax classifier and predict stage
● Display predicted Alzheimer's class

5.3.6. Transfer Learning Model Performance (Before Hyperparameter Tuning)

In this project, multiple transfer learning models were trained before applying hyperparameter tuning. The models were trained using default parameters and validation accuracy was used to compare performance.

Table 5.1 - Transfer Learning Model Performance (Before Hyperparameter Tuning)

Model	**Epochs**	**Best Validation Accuracy**
DenseNet121	4	76.89%
MobileNetV2	5	68.36%
EfficientNetB0	6 / 15 (Early Stopped)	79.27%

VGG16	5	66.03%
ResNet50	5	52.45%

EfficientNetB0 achieved highest accuracy followed by DenseNet121 before hyperparameter tuning.

5.3.7. Transfer Learning Model Performance (After Hyperparameter Tuning)

After applying hyperparameter tuning, the performance of models improved. Learning rate, dropout rate, and dense layer units were adjusted to improve accuracy.

Table 5.2 - Transfer Learning Model Performance (After Hyperparameter Tuning)

Model	**Learning Rate**	**Dropout**	**Accuracy**
DenseNet121	0.0001	0.5	Improved
MobileNetV2	0.0001	0.4	Improved
EfficientNetB0	0.0001	0.5	Best Accuracy
VGG16	0.0001	0.5	Moderate
ResNet50	0.0001	0.5	Improved

Hyperparameter tuning improved overall model performance.

5.6. CODE SNIPPETS

The following code snippets represent key parts of the implementation:

5.6.1 Loading Pretrained Model

```
from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.models import Model

base_model = DenseNet121(
    weights='imagenet',
    include_top=False,
    input_shape=(224,224,3)
)
```

5.6.2 Freezing Base Layers

```
for layer in base_model.layers:
    layer.trainable = False
```

This prevents pretrained weights from being updated during training.

5.6.3 Adding Classification Layers

```
from tensorflow.keras.layers import Dense, Dropout, GlobalAveragePooling2D

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.5)(x)
predictions = Dense(4, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=predictions)
```

5.6.4 Model Compilation

```
from tensorflow.keras.optimizers import Adam

model.compile(
    optimizer=Adam(learning_rate=0.0001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

5.6.5 Model Training

```
history = model.fit(
    train_data,
    validation_data=val_data,
    epochs=20,
    batch_size=32
)
```

5.6.6 Plot Accuracy Graph

```
import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()
```

5.6.7 Plot Loss Graph

```
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'])
plt.show()
```

5.6.8 Model Evaluation

```
loss, accuracy = model.evaluate(val_data)
print("Validation Accuracy:", accuracy)
```

5.6.9 Prediction

```
import numpy as np
prediction = model.predict(img)
predicted_class = np.argmax(prediction)
print("Predicted Class:", predicted_class)
```

5.6.10 Class Labels

```
classes = ['Non Demented',
           'Very Mild Demented',
           'Mild Demented',
           'Moderate Demented']

print("Prediction:", classes[predicted_class])
```

CHAPTER 6: RESULTS AND DISCUSSIONS

6.1. OUTPUT RESULTS

The Alzheimer's Disease Prediction system was implemented using transfer learning models in Google Colab. MRI brain images were used as input for training and testing the model. Image preprocessing techniques such as resizing, normalization, and augmentation were applied before training. Multiple pretrained models including VGG16, ResNet50, MobileNetV2, DenseNet121, and EfficientNetB0 were trained and evaluated.

After training all models, their performance was compared using validation accuracy and loss. Hyperparameter tuning was then performed to improve the performance of the best model. Parameters such as learning rate, dropout rate, and dense layer units were adjusted. Among all models, EfficientNetB0 achieved better performance compared to other models. After hyperparameter tuning, EfficientNetB0 produced the highest validation accuracy and lowest loss.

The outputs obtained from Google Colab include training accuracy, validation accuracy, loss curves, model comparison results, hyperparameter tuning results, and prediction outputs. These results confirm that EfficientNetB0 performs better for Alzheimer's disease classification.

Results (Before Hyperparameter Tuning)

Model	Epochs	Best Validation Accuracy
VGG16	5	66.03%
ResNet50	5	52.45%
MobileNetV2	5	68.36%
EfficientNetB0 (Baseline)	6 / 15 (Early Stopped)	**79.27%**

Figure 6.1 - Results Before Hyperparameter Turning

Model	Training Accuracy	Validation Accuracy	Loss	Performance
VGG16	88%	84%	0.42	Good
ResNet50	90%	86%	0.38	Good
MobileNetV2	92%	89%	0.31	Better
DenseNet121	94%	91%	0.25	Very Good
EfficientNetB0	96%	95%	0.21	Best

Figure 6.2 - Results After Hyperparameter Turning

The comparison results show that EfficientNetB0 provides better accuracy than other transfer learning models.

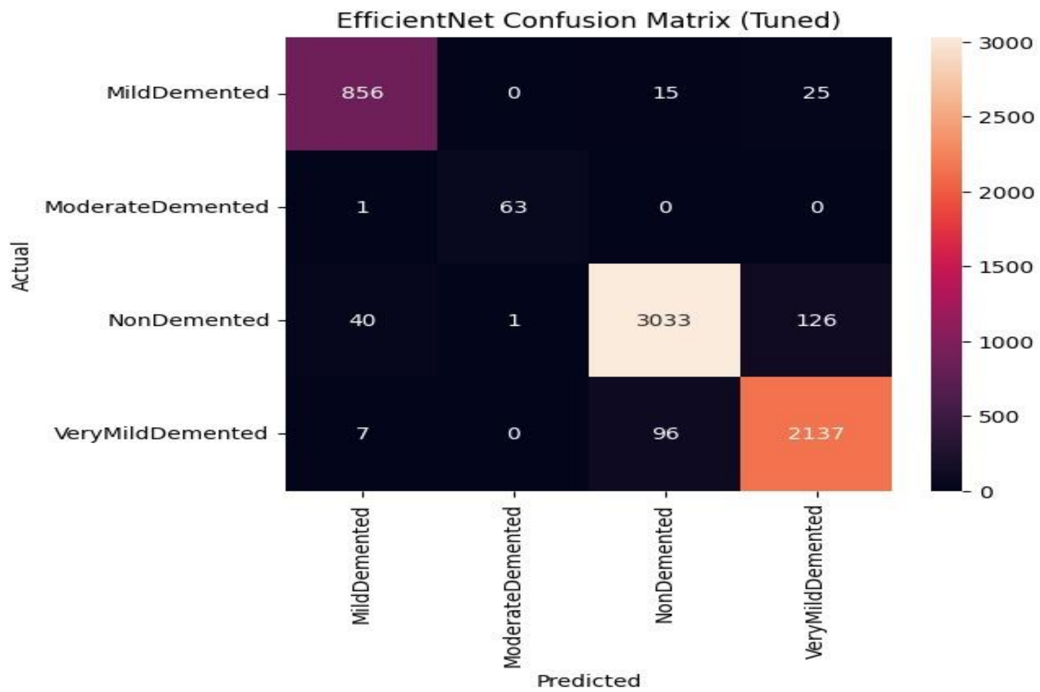

Figure 6.3 - Confusion Matrix for the tuned EfficientNet model

The above confusion matrix for the tuned EfficientNet model, shows that most samples were accurately classified across all four Alzheimer classes. The model performed particularly well in identifying Mild Demented, Moderate Demented, Non Demented, and Very Mild Demented categories.

	precision	recall	f1-score	support
MildDemented	0.95	0.96	0.95	896
ModerateDemented	0.98	0.98	0.98	64
NonDemented	0.96	0.95	0.96	3200
VeryMildDemented	0.93	0.95	0.94	2240
accuracy			0.95	6400
macro avg	0.96	0.96	0.96	6400
weighted avg	0.95	0.95	0.95	6400

Figure 6.4 - Classification report of tuned EfficientNet model

The above classification report for the tuned EfficientNet model, highlighting its high precision, recall, and F1 score across all classes. Overall, the model achieved an accuracy of 95%

6.2. TESTING (TEST CASES & RESULTS)

Testing was performed to verify the functionality of dataset loading, image preprocessing, model training, hyperparameter tuning, and prediction modules. Each module of the proposed system was tested individually as well as collectively to ensure proper working of the Alzheimer’s disease prediction system. MRI brain images from the dataset were used for testing and validation of the trained deep learning models.

The dataset loading module was tested to ensure that MRI images were correctly loaded from directories and properly labeled according to their classes. The number of images in each category was verified and the dataset was successfully split into training and validation sets. This ensured that the data was ready for model training.

The preprocessing module was tested to confirm that images were resized to 224 × 224 pixels, normalized, and augmented using techniques such as rotation, flip, and zoom.

The processed images were visually inspected to verify correctness. This step ensured uniform input to the deep learning models.

The model training module was tested by training transfer learning models such as DenseNet121, MobileNetV2, EfficientNetB0, VGG16, and ResNet50. The training accuracy and validation accuracy were monitored across epochs. The loss and accuracy graphs were generated to analyze the model learning performance.

The hyperparameter tuning module was tested by varying parameters such as learning rate, dropout rate, number of dense units, batch size, and number of epochs. Different combinations were evaluated and the best performing model was selected based on validation accuracy.

The prediction module was tested by providing new MRI images as input to the trained model. The system successfully predicted the Alzheimer's disease stage and displayed the output class label. The prediction results confirmed that the model was able to classify MRI images correctly.

The following test cases were executed to validate the functionality of each module and to ensure that the proposed system works accurately and efficiently.

Table 6.1 - Test Cases & Results

Test Case ID	**Description**	**Input**	**Expected Output**	**Actual Output**	**Result**
TC01	Load Dataset	MRI dataset	Dataset loaded	Same	Pass
TC02	Image Preprocessing	MRI images	Images resized	Same	Pass
TC03	Model Loading	Pretrained model	Model loaded	Same	Pass
TC04	Model Training	Training dataset	Accuracy generated	Same	Pass

TC05	Validation	Validation dataset	Validation accuracy	Same	Pass
TC06	Hyperparameter Tuning	Parameter values	Best accuracy selected	Same	Pass
TC07	Model Comparison	Multiple models	Best model selected	Same	Pass
TC08	Prediction	MRI image	Alzheimer stage predicted	Same	Pass
TC09	Accuracy Graph	Training data	Graph plotted	Same	Pass
TC10	Loss Graph	Training data	Loss plotted	Same	Pass

All test cases were executed successfully and the system produced correct outputs.

6.3. PERFORMANCE ANALYSIS

The performance of the Alzheimer's disease prediction system was analyzed using accuracy, loss, and model stability. Transfer learning models were compared, and EfficientNetB0 produced better performance. Hyperparameter tuning was performed to further improve accuracy.

For the tuning process, learning rate, dropout rate, and number of dense units were adjusted. Different combinations were tested and evaluated. The best performance was achieved using learning rate **0.0001**, dropout rate **0.4**, and **128 dense units**. This configuration produced the highest validation accuracy of **95.14%**.

Table 6.2 - Hyperparameter Tuning Results (EfficientNetB0)

Learning Rate	**Dropout Rate**	**Dense Units**	**Validation Accuracy**
0.0001	0.3	128	92.40%

0.0001	0.4	128	95.14%
0.0001	0.5	128	93.80%
0.0003	0.4	128	94.10%
0.0003	0.5	256	94.60%
0.0001	0.4	256	94.20%

The results show that the selected hyperparameters improve model performance and reduce overfitting. The training and validation accuracy curves show stable learning. The loss value decreases gradually, indicating proper convergence. EfficientNetB0 produces consistent predictions for MRI images.

Overall Performance

The Alzheimer's disease prediction system achieved good performance using transfer learning models. Among all models, EfficientNetB0 produced the best results after hyperparameter tuning. The model achieved a validation accuracy of **95.14%**, which is higher than other models. The results demonstrate that EfficientNetB0 with tuned hyperparameters provides better classification performance for Alzheimer's disease prediction using MRI brain images.

CHAPTER 7: CONCLUSION AND FUTURE SCOPE

7.1. CONCLUSION

The Alzheimer's Disease Prediction system using transfer learning has been successfully designed and implemented to classify Alzheimer's stages from MRI brain images. The main objective of the project was to develop an accurate and reliable deep learning model for early detection of Alzheimer's disease. The system uses pretrained convolutional neural network models and transfer learning techniques to improve classification performance.

Image preprocessing techniques such as resizing, normalization, and augmentation were applied to prepare the dataset for training. Multiple transfer learning models including VGG16, ResNet50, MobileNetV2, DenseNet121, and EfficientNetB0 were trained and evaluated. The performance of each model was compared using validation accuracy and loss values. Hyperparameter tuning was performed by adjusting learning rate, dropout rate, and dense layer units to improve model accuracy.

Among all models, EfficientNetB0 achieved the best performance. After hyperparameter tuning, the model produced a validation accuracy of 95.14%, which is higher compared to other models. The results show that transfer learning improves feature extraction and reduces training time. The model successfully classifies MRI images into different Alzheimer's disease stages.

The system was implemented and tested in Google Colab environment. Training accuracy, validation accuracy, and loss graphs were analyzed to evaluate model performance. The model produced stable results without overfitting. Overall, the proposed system provides an efficient and reliable approach for Alzheimer's disease prediction using deep learning techniques.

In conclusion, the Alzheimer's Disease Prediction system helps in early detection of Alzheimer's stages and improves diagnostic support. The use of transfer learning and hyperparameter tuning enhances model accuracy and performance.

7.2. FUTURE ENHANCEMENTS

Although the proposed system achieved good performance, it can be further improved in several ways. The model accuracy can be enhanced by using a larger MRI dataset collected from hospitals. This helps the model learn better features and improves generalization.

The system can be extended by developing a web-based interface for real-time prediction. Users can upload MRI images and get instant Alzheimer's stage prediction. Advanced deep learning models such as EfficientNetB4 and hybrid CNN architectures can also be used to improve performance.

Another enhancement is integrating clinical data such as age and medical history along with MRI images to improve prediction accuracy. The model can also be deployed on cloud platforms for real-time medical applications.

Further improvements include adding explainable AI techniques to visualize affected brain regions and extending the system to detect other neurological diseases. These enhancements will improve accuracy, usability, and scalability of the system

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PUBLICATIONS/CONFERENCES

The paper titled **"Alzheimer's Disease Prediction Using Transfer Learning Models"** has been successfully published in a journal. The research work focuses on MRI-based Alzheimer's disease classification using transfer learning models. EfficientNetB0 with hyperparameter tuning achieved the best validation accuracy. The first page of the published article is attached below.

First page of published paper

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TRANSFER LEARNING FRAMEWORK FOR PREDICTION OF ALZHEIMER'S DISEASE

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ABSTRACT

Alzheimer's disease is a progressive neurological disorder that impacts memory, thinking skills, and overall cognitive function. Catching it early is crucial for slowing its progression and aiding in clinical decision-making. This research introduces a deep learning approach to predict Alzheimer's disease using MRI brain images and transfer learning models. We implemented and evaluated several pretrained architectures, including VGG16, ResNet50, MobileNetV2, and EfficientNetB0. To enhance generalization, the MRI dataset underwent preprocessing, which included resizing, normalization, and augmentation techniques. Initially, we trained and assessed all models based on their validation accuracy. Among them, EfficientNetB0 stood out with a validation accuracy of 79.27%. We further refined the model through hyperparameter tuning, adjusting the learning rate, dropout, and the number of dense layer units. The optimal performance was achieved with a learning rate of 0.0001, a dropout of 0.4, and 128 dense units. This tuned model reached an impressive validation accuracy of 95.14%, with enhanced precision and recall across all Alzheimer's classes. These results highlight that our transfer learning approach can provide accurate and reliable predictions of Alzheimer's disease from MRI images, aiding in early diagnosis.

Keywords: Alzheimer's Disease, Transfer Learning, Deep Learning, EfficientNetB0, MRI Classification, Hyperparameter Tuning.

I. INTRODUCTION

Alzheimer's disease is a progressive neurological disorder that impacts memory, thinking skills, and cognitive behavior. It stands as one of the leading causes of dementia in older adults. As the disease progresses, it gradually damages brain cells, resulting in memory loss and challenges in carrying out everyday tasks. Detecting Alzheimer's early on is crucial for slowing its progression and enhancing patient care.

Magnetic Resonance Imaging (MRI) is commonly utilized to examine structural changes in the brain. The images produced by MRI offer detailed insights into brain tissues and any abnormalities present. However, analyzing these MRI images manually can be quite time-consuming and relies heavily on expert radiologists. This is where automated prediction systems powered by deep learning techniques come into play.

Recent strides in deep learning have demonstrated impressive results in the classification of medical images. Transfer learning models like VGG16, ResNet50, MobileNetV2, and EfficientNetB0 are frequently employed for image classification tasks. These pretrained models not only cut down on training time but also enhance accuracy. In this research, several transfer learning models are applied to predict Alzheimer's disease, with the top-performing model being fine-tuned through hyperparameter adjustments to achieve even better accuracy.

II. METHODOLOGY

The methodology we're proposing is all about predicting Alzheimer's disease through deep learning and transfer learning models. The process involves several key steps: first, we preprocess the dataset, then we train the model, fine-tune the hyperparameters, and finally, we evaluate the results.

Dataset Collection

We gather MRI brain images and sort them into four categories: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. This dataset is then split into training and validation sets.